

ALKANE REACTIONS

NAME	REACTANTS	CONDITIONS	PRODUCTS
Cracking	Alkanes only	Heat with aluminum oxide catalyst	Alkanes and alkenes
Chlorination or Bromination (Free radical substitution)	Alkanes with chlorine or bromine	Presence of UV light (sunlight)	Alkyl halide
Combustion	Alkanes with oxygen	Burn	Carbon dioxide and water

ALKENE REACTIONS

NAME	REACT WITH	CONDITIONS	PRODUCTS
Hydrogenation (Electrophilic addition)	Hydrogen	Heat with platinum or nickel catalyst	Alkane
Electrophilic addition	Steam	Heat with phosphoric acid catalyst	Alcohol
Electrophilic addition	Hydrogen halide	RTP	Alkyl halide
Electrophilic addition	Halogen molecules	RTP	Alkyl halide
Oxidation	Cold dilute acidified potassium manganate		Diol, purple solution goes colorless
Oxidation	Hot concentrated acidified potassium manganate with =CH ₂		Carbon dioxide
Oxidation	Hot concentrated acidified potassium manganate with =CH		Aldehyde and then carboxylic acid
Oxidation	Hot concentrated acidified potassium manganate with =C		Ketone
Additional polymerization	Other alkenes		Long chain polymer

ALKYL HALIDE REACTIONS

NAME	REACT WITH	CONDITIONS	PRODUCTS
Nucleophilic substitution	Aqueous NaOH (or other aqueous alkali)	Heat aqueous solution	Alcohol
Nucleophilic substitution	KCN	Ethanolic solution heated under reflux	Nitrile
Nucleophilic substitution	Ammonia	Ethanolic solution of aqueous ammonia heated under pressure	Primary amine
Hydrolysis	Aqueous silver nitrate	In ethanol	Alcohol and silver halide
Elimination	NaOH	Ethanolic solution heated	Alkene, salt, and water

ALCOHOL REACTIONS

NAME	REACTANTS	CONDITIONS	PRODUCTS
Combustion	Oxygen	Ignited	Carbon dioxide and water
Nucleophilic substitution	HX (KBr in sulfuric or phosphoric acid is used instead of HBr)		Alkyl halide and water
	PCl ₃	Heat	Alkyl halide and H ₃ PO ₃
	PCl ₅	RTP	Alkyl halide, hydrochloric acid, and POCl ₃
	Sodium		Hydrogen and salt of alcohol
Oxidation	Primary alcohol with potassium manganate or potassium dichromate	Acidic solution, warm alcohol	Aldehyde and then carboxylic acid

Oxidation	Secondary alcohol with potassium manganate or potassium dichromate	Acidic solution, warm alcohol	Ketone
Dehydration	Acid, aluminum oxide, porous pot, or pumice	Concentrated if acid, heated if other	Alkene and water
Esterification	Carboxylic acid	Heated under reflux with strong acid catalyst	Ester

CARBONYL REACTIONS

NAME	REACTANTS	CONDITIONS	PRODUCTS
Reduction	Aldehyde with NaBH ₄ or LiAlH ₄		Primary alcohol
Reduction	Ketone with NaBH ₄ or LiAlH ₄		Secondary alcohol
Nucleophilic addition	HCN	KCN catalyst and heat	Hydroxynitrile
Condensation or Carbonyl group test	2,4-DNPH	Heat	Water and deep orange precipitate
Oxidation	Aldehyde with Fehling’s solution	Warmed in alkaline solution	Salt of acid and red copper oxide
Oxidation	Aldehyde with Tollen’s reagent	Warmed in alkaline solution	Salt of acid and silver mirror of Ag atoms

CARBOXYLIC ACID REACTIONS

NAME	REACTANTS	CONDITIONS	PRODUCTS
Redox	Reactive metals like sodium		Salt of acid and hydrogen
Neutralization	Alkali		Salt of acid and hydrogen
Acid-base	Carbonates		Salt, water, and carbon dioxide
Esterification	Alcohols	Concentrated sulfuric acid catalyst and heated under reflux	Ester and water
Reduction	LiAlH ₄		Primary alcohol and water

ESTER REACTIONS

NAME	REACTANTS	CONDITIONS	PRODUCTS
Hydrolysis (reversible)	Dilute acid	Heat under reflux	Carboxylic acid and alcohol
Hydrolysis (irreversible)	Dilute alkali	Heat under reflux	Salt of acid and alcohol. Salt becomes carboxylic acid after acidification

NITRILE REACTIONS

NAME	REACTANTS	CONDITIONS	PRODUCTS
Hydrolysis	Dilute acid	Heat	Carboxylic acid and ammonium salt
Hydrolysis	Dilute alkaline	Heat	Salt of acid and ammonia. Salt can be converted to acid through acidification.

ALKANE PREPARATION

NAME	REACTANTS	CONDITIONS	PRODUCTS
Hydrogenation (Addition)	Alkene and hydrogen	Heated over fine platinum or nickel catalyst	Alkane
Cracking	Long chain alkane	Heated with aluminum oxide catalyst	Alkanes and alkenes

ALKENE PREPARATION

NAME	REACTANTS	CONDITIONS	PRODUCTS
Elimination	Alkyl halide with NaOH	Ethanolic solution of NaOH, heat	Alkene, water, and NaX
Dehydration	Alcohol with acid, or aluminum oxide/pot/pumice	Concentrated if acid, hot if other	Alkene and water
Cracking	Long hydrocarbon	Heat with aluminum oxide catalyst	Alkenes and alkanes

ALKYL HALIDE PREPARATION:

NAME	REACTANTS	CONDITIONS	PRODUCTS
Free radical substitution of alkanes	Alkane with chlorine or bromine	Presence of UV light	Alkyl halide, HX, alkanes
Electrophilic addition of alkenes	Alkene with HX or X ₂	RTP	Alkyl halide
Substitution of alcohol	Alcohol with : 1) HX 2) KBr in phosphoric or sulfuric acid 3) PCl ₃ 4) PCl ₅ 5) SOCl ₂	Heat if PCl ₃ , RTP if PCl ₅	Alkyl halide and: 1) Water 2) Water 3) H ₃ PO ₃ 4) HCl and POCl ₃ 5) HCl and SO ₂

ALCOHOL PREPATION

NAME	REACTANTS	CONDITIONS	PRODUCTS
Electrophilic addition of alkenes	Alkene with steam	Heat with phosphoric acid catalyst	Alcohol
Oxidation of alkene	Alkene with old dilute acidified potassium manganate		Diol, purple solution goes colorless
Subsitution of halogenoalkane	Alkyl halide with aqueous NaOH (or other aqueous alkali)	Aqueous	Alcohol
Reduction	Aldehyde with NaBH ₄ or LiAlH ₄		Primary alcohol
Reduction	Ketone with NaBH ₄ or LiAlH ₄		Secondary alcohol
Reduction	Carboxylic acid with LiAlH ₄		Primary alcohol and water
Hydrolysis (reversible)	Ester with dilute acid	Heat under reflux	Carboxylic acid and alcohol
Hydrolysis (irreversible)	Ester with dilute alkali	Heat under reflux	Salt of acid and alcohol. Salt becomes carboxylic acid after acidification

CARBONYL PREPARATION

NAME	REACTANTS	CONDITIONS	PRODUCTS
Oxidation	Primary alcohol with potassium manganate or potassium dichromate	Acidic solution, warm alcohol. Product is distilled immediately	Aldehyde
Oxidation	Secondary alcohol with potassium manganate or potassium dichromate	Acidic solution, warm alcohol	Ketone

CARBOXYLIC ACID PREPARATION

NAME	REACTANTS	CONDITIONS	PRODUCTS
Oxidation	Primary alcohol with potassium manganate or potassium dichromate	Acidic solution, warm alcohol.	Carboxylic acid
Hydrolysis	Nitrile with dilute acid	Heat	Carboxylic acid and ammonium salt
Hydrolysis	Ntrile with dilute alkali	Heat	Salt of acid and ammonia. Salt can

			be converted to acid through acidification
Hydrolysis (reversible)	Ester with dilute acid	Heat under reflux	Carboxylic acid and alcohol
Hydrolysis (irreversible)	Ester with dilute alkali	Heat under reflux	Salt of acid and alcohol. Salt becomes carboxylic acid after acidification

ESTER PREPARATION

NAME	REACTANTS	CONDITIONS	PRODUCTS
Esterification	Carboxylic acid	Heated under reflux with strong acid catalyst	Ester

PRIMARY AMINE PRODUCTION

NAME	REACTANTS	CONDITIONS	PRODUCTS
Nucleophilic substitution	Alkyl halide with ammonia	Ethanolic solution of ammonia heated under pressure	Primary amine

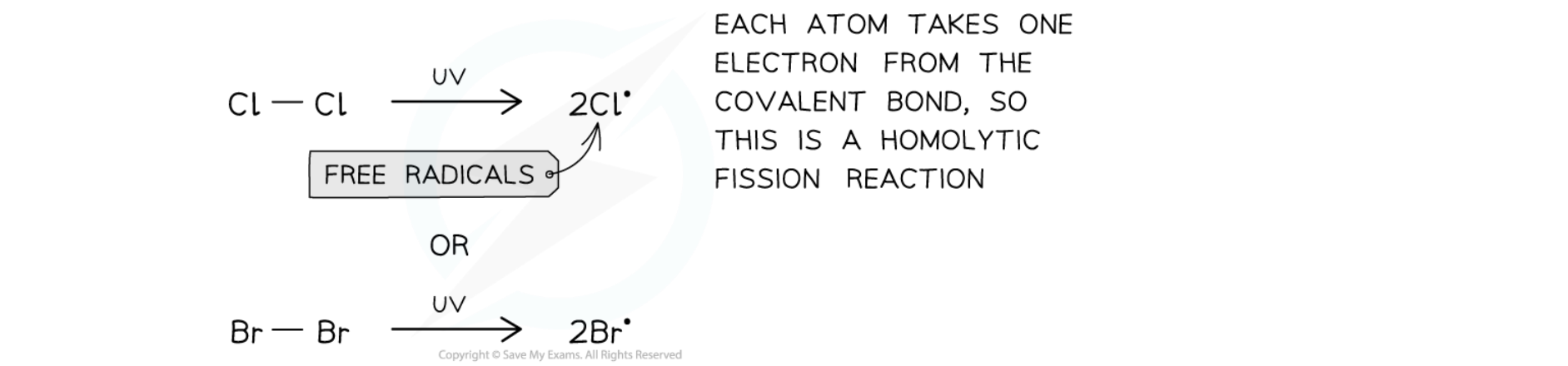
NITRILE PRODUCTION

NAME	REACTANTS	CONDITIONS	PRODUCTS
Nucleophilic substitution	Alkyl halide with KCN	Ethanolic solution heated under reflux	Nitrile

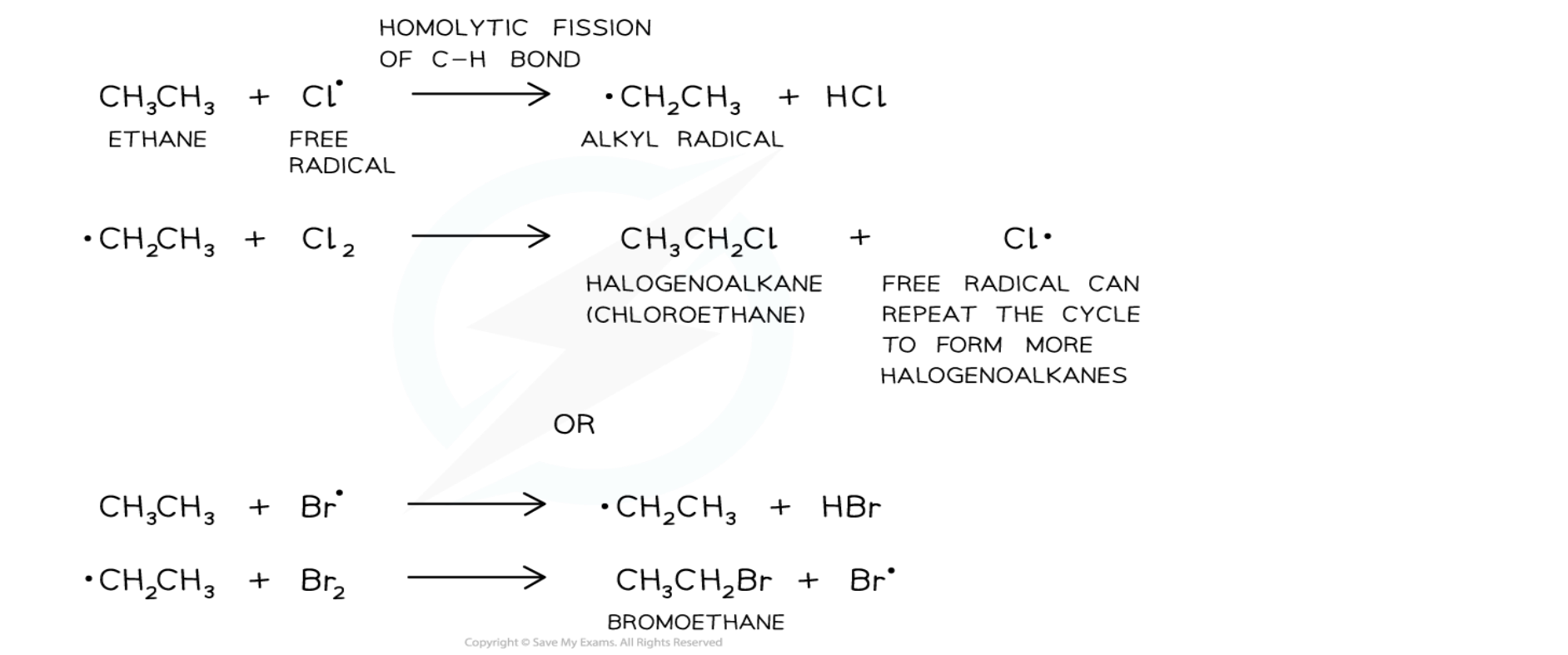
MECHANISMS

Free Radical Substitution of Alkanes:

INITIATION STEP: Homolytic fission of halogen molecule to form free radicals



PROPAGATION STEP: These free radicals attack alkanes or other halogen molecules and turn them into free radicals. The cycle is then repeated.



TERMINATION STEP: Two free radicals react with each other to form a single unreactive molecule.



THE SAME PROCESS CAN TAKE PLACE WITH BROMINE INSTEAD OF CHLORINE

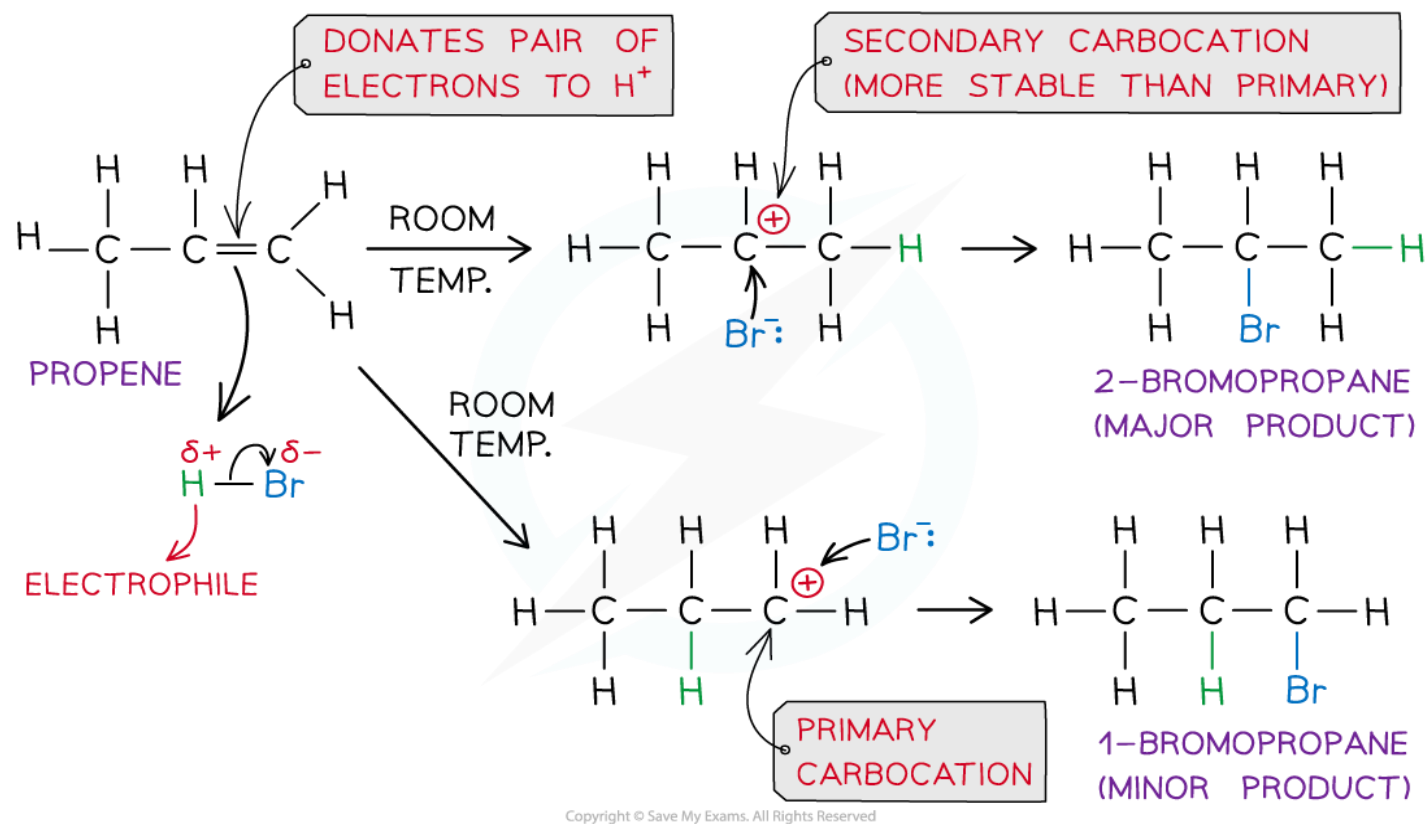
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Electrophilic Addition of Alkenes:

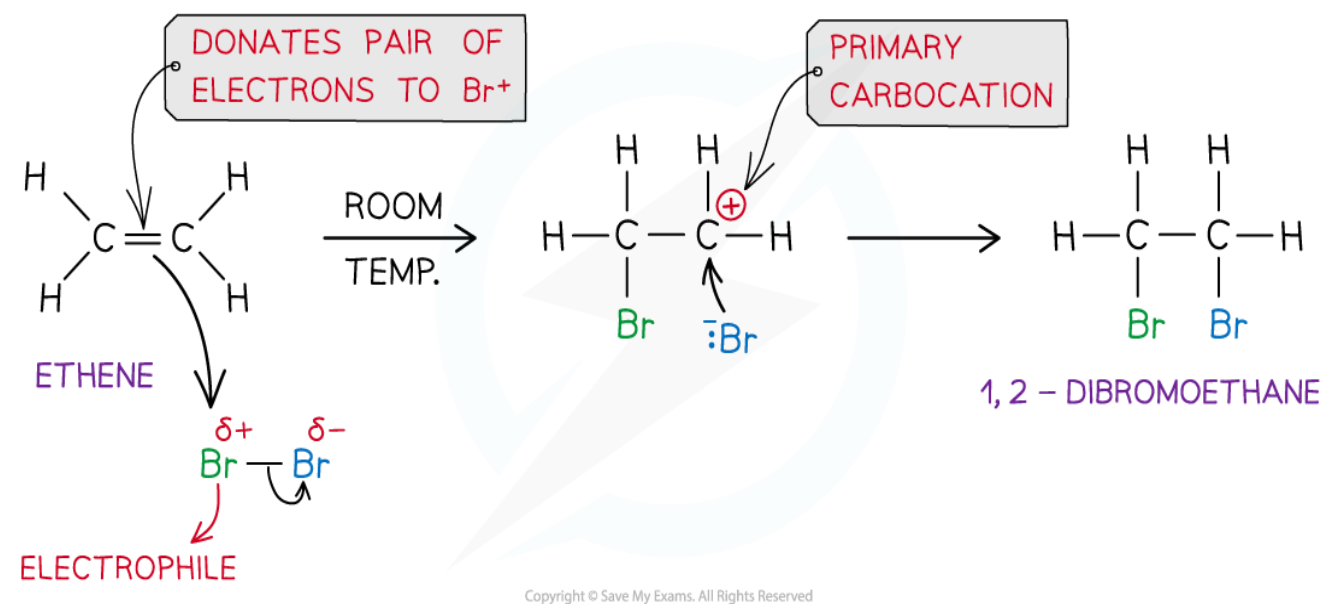
Step 1: The H-Br bond breaks heterolytically, forming a Br^- ion and a H^+ ion because bromine has a higher electronegativity.

Step 2: The H^+ ion attacks the organic compound and takes a pair of electrons from it, leaving a highly reactive carbocation which is the intermediate compound.

Step 3: The bromide electrophile donates a pair of electrons and reacts with the carbocation to form an alkyl halide.



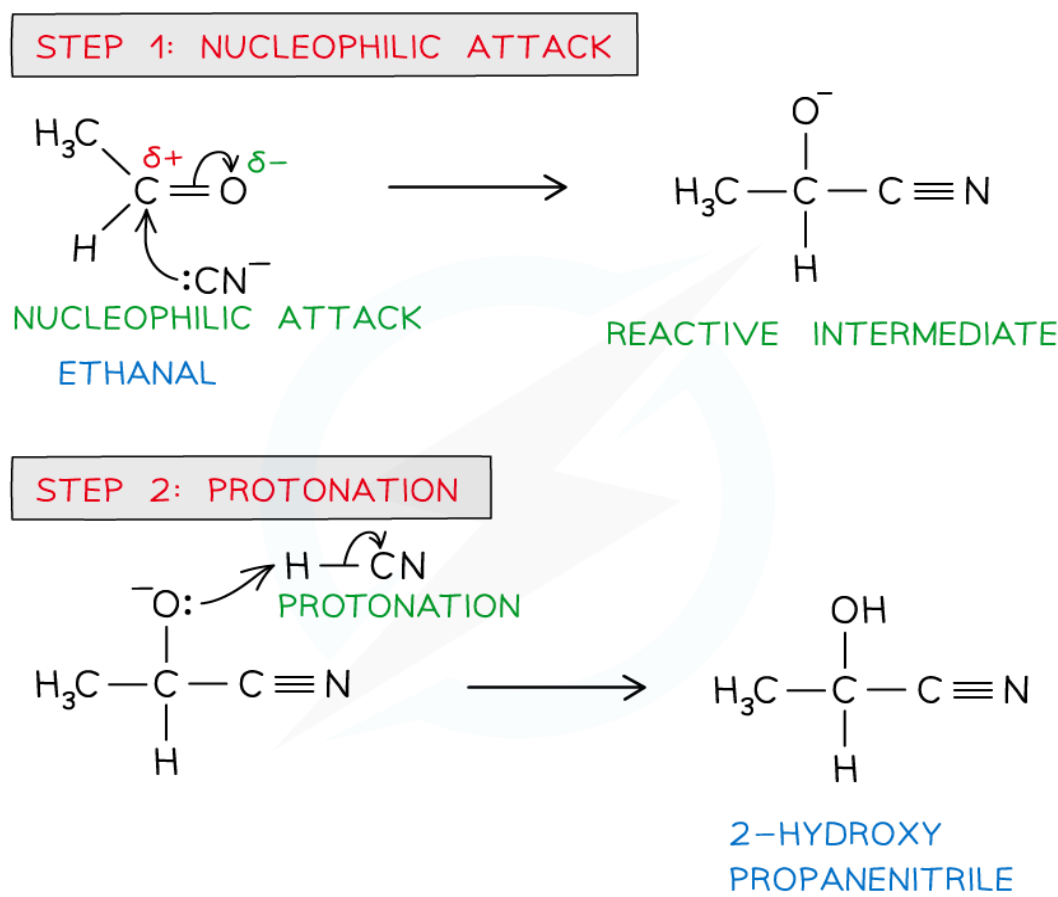
The same process takes place when nonpolar halogen molecules react with alkenes, with one small difference: they have to be polarized first before heterolytic fission can occur. When the halogen molecule comes close to the $\text{C}=\text{C}$ bond, the halogen atom closer to the $\text{C}=\text{C}$ bond gets a partial positive charge and the atom further away gets a partial negative charge, hence, heterolytic fission occurs.



Nucleophilic Addition of HCN with Carbonyl Compounds:

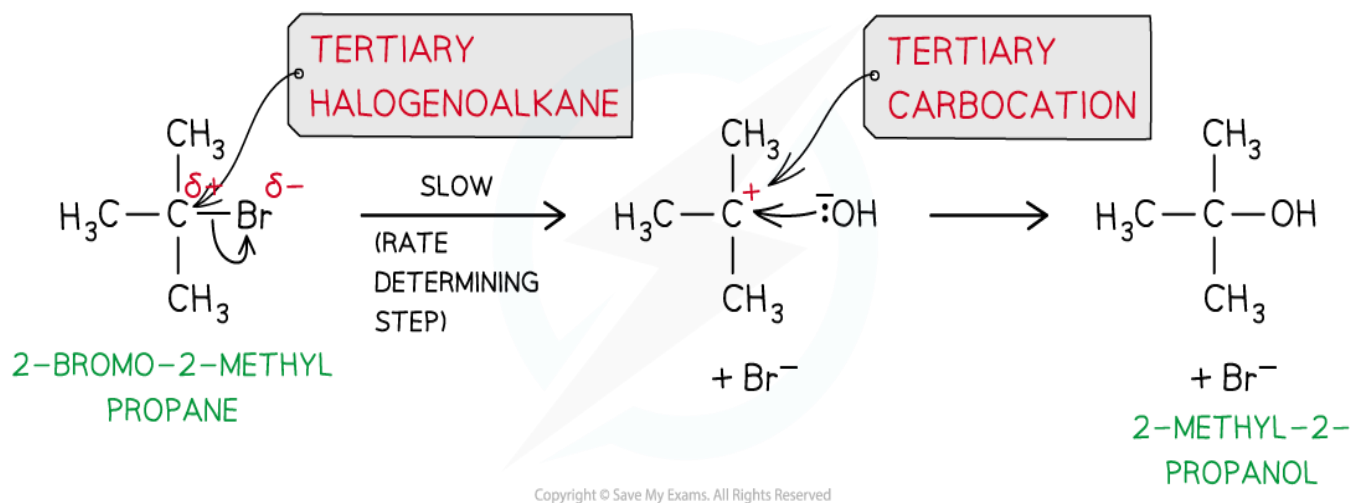
Step 1: Cyanide ion attacks carbonyl atom to form a negatively charged intermediate

Step 2: The negatively charged oxygen atom in the reactive intermediate quickly reacts with aqueous H^+ (either from HCN, water or dilute acid) to form 2-hydroxynitrile

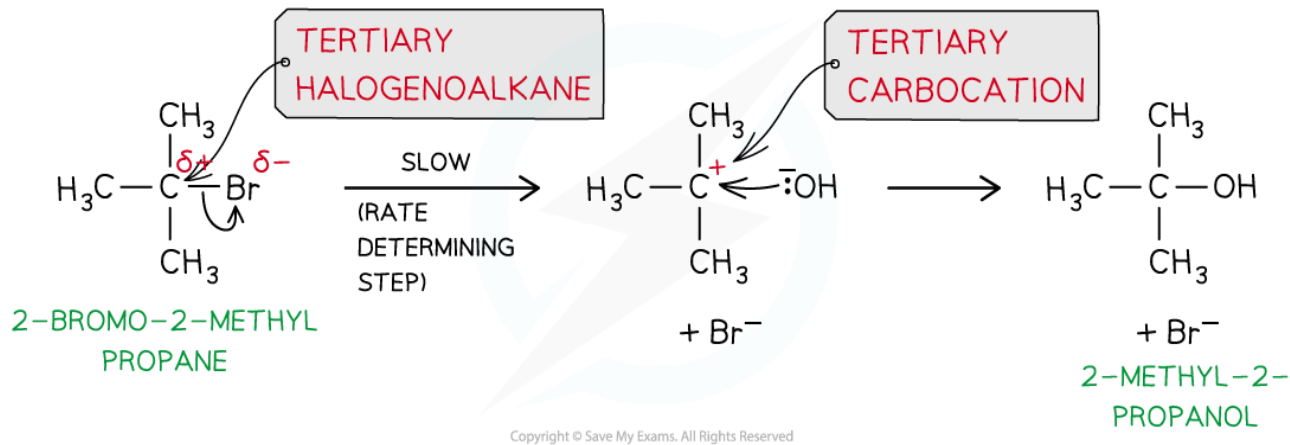


SN1 MECHANISM (Tertiary Alkyl Halides):

Step 1: C-X bond breaks **heterolytically** and the halogen leaves the halogenoalkane as an X⁻ ion (this is the **slow** and **rate-determining** step)

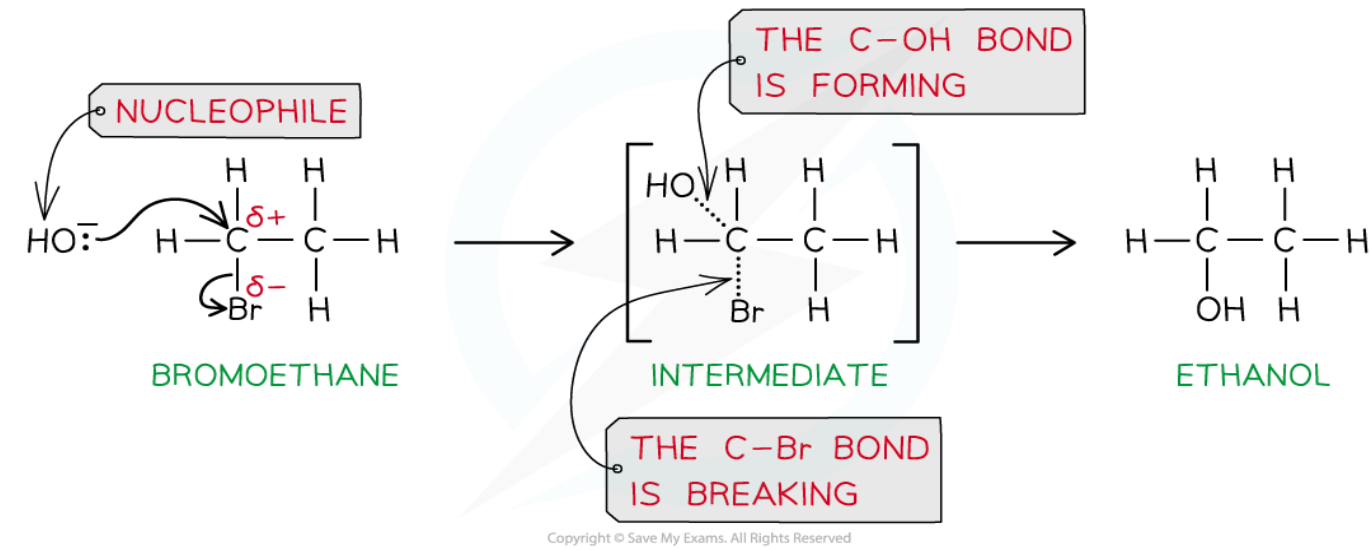


Step 2: Tertiary carbocation is attacked by the nucleophile



SN2 MECHANISM (Primary Alkyl Halides):

The nucleophile donates a pair of electrons to the δ^+ carbon atom to form a new bond. At the same time, the C-X bond is breaking and the halogen (X) takes both electrons in the bond (heterolytic fission). The halogen leaves the halogenoalkane as an X⁻ ion.



ALKANES MISCELLANEOUS:

Chemical Properties:

- a) nonpolar compounds due to negligible difference of electronegativity between hydrogen and carbon.
- b) do not react with polar compounds because they have no electron deficient areas to attract nucleophiles nor any electron rich areas to attract electrophiles.
- c) unreactive at rtp due to strong C-C and C-H bonds.

Obtaining from Crude Oil:

- a) Fractional distillation to separate similar hydrocarbons first.
- b) Crack excess heavier hydrocarbons into smaller more useful compounds.

Effects of Pollutants from Alkanes:

	Formation	Environmental Consequence	Catalytic Removal
Carbon Monoxide	Incomplete combustion of alkanes in car engines	Toxic gas	Oxidation to CO ₂ : $2\text{CO} + \text{O}_2 \longrightarrow 2\text{CO}_2$ OR $2\text{CO} + 2\text{NO} \longrightarrow 2\text{CO}_2 + \text{N}_2$
Oxides of Nitrogen	Oxidation of nitrogen in car engines	Dissolve in and react in water with oxygen to form acid rain	Reduction to nitrogen gas: $2\text{CO} + 2\text{NO} \longrightarrow 2\text{CO}_2 + \text{N}_2$
VOCs	Unburnt hydrocarbons from fuels and their oxides formed in car engines	React with oxides of nitrogen in the atmosphere to produce PAN	Oxidise unburnt hydrocarbons to carbon dioxide and water: $\text{C}_n\text{H}_{2n+2} + (3n + 1)[\text{O}] \longrightarrow n\text{CO}_2 + (n + 1)\text{H}_2\text{O}$
PAN	From the photochemical reaction of VOCs and nitrogen oxides in the atmosphere	Photochemical smog	Oxidise unburnt hydrocarbons and reduce nitrogen oxides to prevent formation of PAN in the atmosphere

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Carbon monoxide is toxic. It binds to hemoglobin in blood, hence stopping oxygen from circulating around the body, leading to dizziness, unconsciousness, and possible death. Carbon monoxide is odorless and hence can't be detected by humans.

Acid rain can corrode buildings, damage plants and wildlife, and harm human health.

ALKENES MISCELLANEOUS:

Markovnikov’s Rule:

- 1) The most stable carbocations are tertiary, then secondary, then primary. This is because tertiary carbocations have three electron donating alkyl groups attached to the positively charged carbon, reducing its positive charge and hence making it more stable.
- 2) During electrophilic addition, the electrophile will react with the carbon which gives the most stable carbocation.
- 3) Hence, the electrophile will bind to the carbon with the most alkyl groups attached to it.

ALKYL HALIDES MISCELLANEOUS:

Reaction with Aqueous Silver Nitrate In Ethanol:

- 1) Water reacts with the alkyl halide to form an alcohol and HX.
- 2) HX and silver nitrate react to form AgX precipitate and nitric acid.
- 3) If X is chlorine, ppt is white. If X is bromine, ppt is offwhite/creamy. If X is iodine, ppt is yellow.
- 4) The amount of time it takes ppt to form shows the reactivity of various alkyl halides too. The quicker the ppt forms, the more reactive the alkyl halide is, and the less the bond energy of C-X.
- 5) From most to least reactive: iodoalkanes, bromoalkanes, chloroalkanes, fluoroalkanes.
- 6) Hence C-I is the weakest bond and C-F is the strongest.

S _N 1 Mechanism	S _N 2 Mechanism
Tertiary alkyl halides	Primary alkyl halides
Two steps	One step
Rate of reaction depends on concentration of alkyl halide only	Rate of reaction depends on concentration of both alkyl halide and nucleophile
Carbocation forms	No carbocations formed
Polar solvent	Non polar solvent

ALCOHOLS MISCELLANEOUS:

Why Alcohols Have Less Acidity Than Water:

- 1) Alcohols have a low degree of dissociation in water.
- 2) ROH (aq) ⇌ RO⁻ (aq) + H⁺ (aq)
- 3) The position of equilibrium lies to the left, meaning there are far more atoms of alcohol, so low acidity.
- 3) In the dissociation of water, the position of equilibrium still lies to the left, but there are still more H⁺ ions than in alcohols.

4) The O- atom in an alcohol is bound to an electron donating alkyl group which has a positive inductive effect on oxygen, meaning it increases its electron density and hence increases its negativity. Hence, it is likely to react with H⁺ ions to form alcohols again. Water does not have anything similar, so it does not react with H⁺ ions so often.

Iodoform Test to Identify Position of OH Group:

- 1) Reaction with iodine in alkaline solution.
- 2) If the -OH group is on the carbon atom next to a methyl group, it will firstly get oxidised to CH₃CH(OH)- by the alkaline solution.
- 3) This will result in the formation of a methyl ketone RCOCH₃.
- 4) The methyl ketone will then first get halogenated and then hydrolysed to form the sodium salt and the yellow precipitate.
- 5) If no yellow precipitate is formed, then this means that the secondary alcohol is not on a carbon next to a methyl group.

Additional Polymerization:

- 1) Many monomers containing at least one C=C bond react to form a long chain polymer
- 2) The π-bond in each C=C bond breaks and then the monomers link together to form new C-C single bonds

Disposal of Polymers:

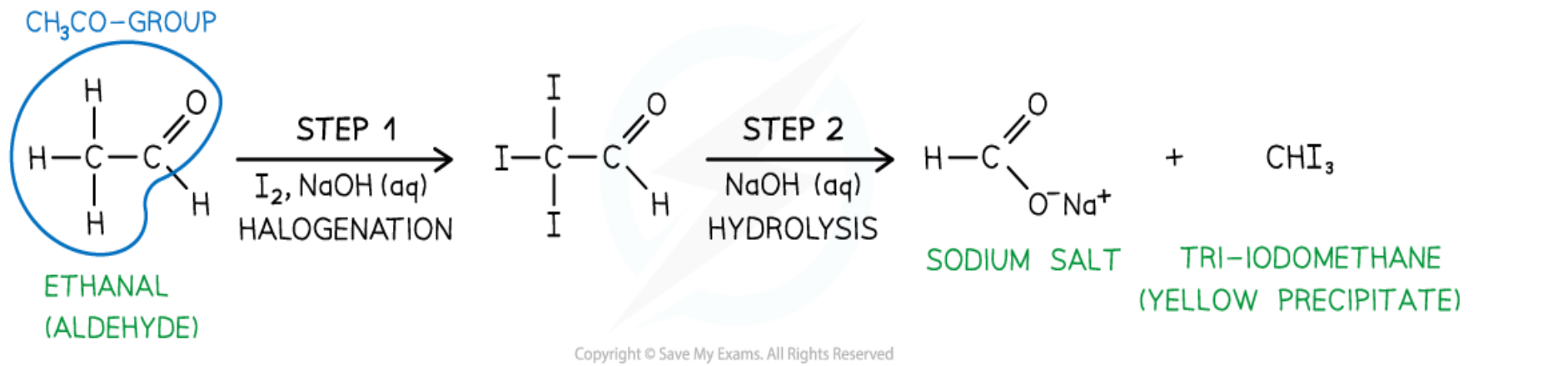
- 1) Burning polymers produces harmful combustion products which pollute the environment.
- 2) They are very large alkanes which are unreactive and hence non-biodegradable.
- 3) Hence they take centuries to decompose in landfills, causing long term pollution of the environment.

TESTS FOR ORGANIC COMPOUNDS

TEST FOR	TEST NAME	TEST REAGENTS AND CONDITIONS	TEST PRODUCTS	POSITIVE TEST OBSERVATIONS
Saturation (C=C bond)		Alkene with bromine water in absence of UV light		Bromine water decolorized
Halide ion		Alkyl halide with aqueous silver nitrate in ethanol		White ppt if chlorine, off-white/creamy ppt if bromine, yellow ppt if iodine
OH group		Sodium with organic compound	0.5 moles of hydrogen gas per OH group and salt of acid/alcohol	Fizzing and bubbles which make pop sound with lighted splinter
OH group		PCl ₅ with compound	Alkyl halide, HCl, and POCl ₃	Misty white fumes of HCl
Tertiary alcohols		K ₂ Cr ₂ O ₇ with alcohol		No change, orange solution stays orange
CH ₃ CO group (methyl ketones, ethanal, some secondary alcohols)	Iodoform	Iodine with compound in alkaline solution heated	Yellow ppt of tri-iodomethane and salt of acid	Yellow ppt
Carbonyl group		2,4-DNPH with compound		Deep orange ppt
Aldehyde	Tollen’s reagent	Silver nitrate in excess ammonia with carbonyl compound	Salt of carboxylic acid and silver atoms	Silver mirror due to silver atoms
Aldehyde	Fehling’s solution	Alkaline solution of copper ions with carbonyl compound	Salt of carboxylic acid and copper (I) oxide	Opaque red ppt of copper (I) oxide
Carboxylic acid		Magnesium with compound	Salt of acid and hydrogen	Fizzing and bubbles which make pop sound with lighted splinter

Iodoform Test for Carbonyl Compounds:

- 1) Test for CH₃CO group (methyl ketones and ethanal).
- 2) Compound is reacted with iodine in heated alkaline solution.
- 3) All three H-atoms in the -CH₃ (methyl) group are replaced with iodine atoms, forming a -CI₃ group.
- 4) The intermediate compound is hydrolysed by the alkaline solution to form a sodium salt (RCO₂⁻ Na⁺) and a yellow precipitate of CHI₃.



Properties of Oxides:

- 1) They are all basic except for BeO which is amphoteric.
- 2) Alkalinity increases down the group.
- 3) Form salt when reacted with HCl and sulphate when reacted with sulfuric acid.
- 4) When an insoluble sulfate is formed during reaction with sulfuric acid, the sulfate forms at the surface of the oxide, preventing the solid oxide underneath from reacting. This can be avoided by using a fine oxide and stirring.

Properties of Reactions with Acids:

- 1) Form colorless solution of metal salts.
- 2) Solubility of sulphates decreases down the group. Barium and strontium sulphate are insoluble, while calcium sulphate is partially soluble.
- 3) Barium sulfate is insoluble white ppt.

Properties of Hydroxides:

- 1) Form salt and water when reacted with acid.
- 2) Solubility and alkalinity increase down the group.

Properties of Carbonates:

- 1) All carbonates are insoluble except for beryllium carbonate.
- 2) Form colorless chlorides, carbon dioxide, and water when reacted with acids.
- 3) Form sulfates with sulfuric acid.
- 4) The carbonates of Ca, Sr and Ba form as an insoluble sulfate layer on their solid carbonates which stops any further reaction after the initial bubbling (effervescence) of carbon dioxide gas is seen.

Thermal Decomposition of Carbonates:

- 1) General equation is $MCO_3(s) \rightarrow MO(s) + CO_2(g)$
- 2) Stability increases down the group, so more heat is needed down the group.

Thermal Decomposition of Nitrates:

- 1) General equation is $2M(NO_3)_2(s) \rightarrow 2MO(s) + 4NO_2(g) + O_2(g)$
- 2) Brown fumes visible due to nitrogen dioxide.
- 3) Thermal stability increases down the group. This is because the smaller positive ions at the top will polarize the nitrate ion more than the larger ions at the bottom, and the more polarized they are, the likelier they are to decompose.

Physical Properties:

- 1) Atomic radius increases as new principal quantum shells are added.
- 2) Melting point decreases down the group as the distance and consequently the attraction between the nucleus and the bonding electrons decreases.
- 3) Density increases down the group.

Chemical Properties:

- 1) Valence shell configuration is ns^2
- 2) Form ions with a 2+ charge to form ionic bonds.
- 3) Reactivity increases down the group because first and second ionization energies decrease down the group.
- 4) Nuclear charge increases, but is offset by increase in shielding effect and atomic radius, so ionization energies decrease down the group.
- 5) Barium is so reactive it must be stored in oil to prevent it from reacting with air.

PERIOD 3

Physical Properties:

- 1) Atomic radius decreases along the group because nuclear charge increases but shielding effect remains constant
- 2) The ionic radius of the cations from Group 1 to Group 14 decreases along the period. This is because nuclear charge increases along the period. As the ions have lost their valence electrons, there is less shielding effect as well, meaning cations are smaller than their atoms.
- 3) The anions are larger than the original atoms because they gain electrons, increasing repulsions in their valence shell while nuclear charge is constant, causing the electron cloud to expand.
- 4) The radius of the anions decreases from Group 15 to Group 17 because nuclear charge increases and gain in electrons decreases.
- 5) Electronegativity increases along the period.
- 6) Melting point increases from Group 1 to Group 14, decreases significantly in Group 15, increases slightly in Group 16, and decreases thereafter. Silicon has the highest melting point.
- 7) This is because from Group 1 to Group 13, the strength of the metallic bonds increases due to increase in valence electrons, leading to more delocalized electrons in the metallic bond.
- 8) Silicon has the highest melting point because it exists in a giant molecular structure where each atom is joined to others with strong covalent bonds.
- 9) The remaining are simple covalent molecules with weak instantaneous dipole-induced dipole forces. Sulfur has a higher melting point than phosphorus because it has eight atoms within a molecule whereas phosphorus only has four, so sulfur has stronger intermolecular instantaneous dipole-induced dipole forces.

- Chlorine has diatomic molecules and argon exists as one atom, so intermolecular forces decrease along the period, and so does melting point.
- 10) Electrical conductivity increases from Group 1 to Group 13 and thereafter decreases across the period. Chlorine and argon do not conduct electricity.
- 11) There is an increase in valence electrons and hence increase in sea of delocalized electrons from sodium to aluminum, so electrical conductivity increases.
- 12) The remaining atoms have no delocalized electrons due to their structure, and hence are bad electrical conductors.

Period 3 Element	Na	Mg	Al	Si	P	S	Cl	Ar
Bonding	Metallic	Metallic	Metallic	Covalent	Covalent	Covalent	Covalent	–
Structure	Giant metallic	Giant metallic	Giant metallic	Giant molecular	Simple molecular	Simple molecular	Simple molecular	Simple molecular

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Reactions with Oxygen:

	Chemical Equation	Reaction Conditions	Reaction	Flame	Product
Na	$4\text{Na(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{Na}_2\text{O(s)}$	Heated	Vigorously	Bright yellow flame	White solid
Mg	$2\text{Mg(s)} + \text{O}_2\text{(g)} \rightarrow 2\text{MgO(s)}$	Heated	Vigorously	Bright white flame	White solid
Al	$4\text{Al(s)} + 3\text{O}_2\text{(g)} \rightarrow 2\text{Al}_2\text{O}_3\text{(s)}$	Powdered Al	Fast	Bright white flame	White powder
Si	$\text{Si(s)} + \text{O}_2\text{(g)} \rightarrow \text{SiO}_2\text{(s)}$	Powdered Si Heat strongly	Slowly	Bright white sparkles	White powder
P	$4\text{P(s)} + 5\text{O}_2\text{(g)} \rightarrow \text{P}_4\text{O}_{10}\text{(s)}$	Heated	Vigorously	Yellow or white flame	White clouds
S	$\text{S(s)} + \text{O}_2\text{(g)} \rightarrow \text{SO}_2\text{(g)}$	Powdered S is heated	Gently	Blue flame	Toxic fumes

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Reactions with Chlorine:

	Chemical Equation	Reaction Conditions	Reaction
Na	$2\text{Na(s)} + \text{Cl}_2\text{(g)} \rightarrow 2\text{NaCl(s)}$	Heated	Vigorously
Mg	$\text{Mg(s)} + \text{Cl}_2\text{(g)} \rightarrow \text{MgCl}_2\text{(s)}$	Heated	Vigorously
Al	$2\text{Al(s)} + 3\text{Cl}_2\text{(g)} \rightarrow \text{Al}_2\text{Cl}_6\text{(s)}$	Heated	Vigorously
Si	$\text{Si(s)} + 2\text{Cl}_2\text{(g)} \rightarrow \text{SiCl}_4\text{(l)}$	Heated	Slowly
P	$2\text{P(s)} + 5\text{Cl}_2\text{(g)} \rightarrow 2\text{PCl}_5\text{(l)}$	Heated Excess chlorine	Slowly

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Reactions with Water:

- Sodium reacts vigorously with cold water.
- $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$
- The sodium melts into a ball and moves across the water surface before disappearing. Hydrogen gas is given off.
- The resulting solution has pH 14 due to the strongly alkaline sodium hydroxide formed.
- Magnesium reacts extremely slowly with cold water
- $\text{Mg(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Mg(OH)}_2\text{(aq)} + \text{H}_2\text{(g)}$
- The solution has pH 11 because magnesium hydroxide is only slightly soluble.
- Magnesium reacts vigorously when heated with steam.
- $\text{Mg(s)} + \text{H}_2\text{O(g)} \rightarrow \text{MgO(s)} + \text{H}_2\text{(g)}$

Physical Properties of Period 3 Oxides:

Period 3 oxide	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀	SO ₂ , SO ₃
Relative melting point	High	High	Very high	Very high	Low	Low
Chemical bonding	Ionic	Ionic	Ionic (with some degree of covalent	Covalent	Covalent	Covalent
Structure	Giant ionic	Giant ionic	Giant ionic	Giant covalent	Simple molecular	Simple molecular

Oxidation States in Oxides:

- Oxygen is more electronegative than all the Period 3 elements.
- Therefore, all the Period 3 elements have a positive oxidation state in their oxides while oxygen has an oxidation state of −2.

Period 3 Oxides With Water:

Oxide	Chemical equation	pH	Comments
Na ₂ O	Na ₂ O(s) + H ₂ O(l) → 2NaOH(aq)	12–14 (strongly alkaline)	–
MgO	MgO(s) + H ₂ O(l) → Mg(OH) ₂ (aq)	8–10 (weakly alkaline)	–
Al ₂ O ₃	No reaction	–	Al ₂ O ₃ is insoluble in water
SiO ₂	No reaction	–	SiO ₂ is insoluble in water
P ₄ O ₁₀	P ₄ O ₁₀ (s) + 6H ₂ O(l) → 4H ₃ PO ₄ (aq)	3–4 (weakly acidic)	Vigorous reaction
SO ₂ , SO ₃	SO ₂ (g) + H ₂ O(l) → H ₂ SO ₃ (aq) SO ₃ (g) + H ₂ O(l) → H ₂ SO ₄ (aq)	1–2 (strongly acidic)	–

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Acid/Base Behavior of Period 3 Oxides:

- All metal oxides except aluminum oxide are alkaline
- Aluminum oxide is amphoteric
- Oxides of Groups 14, 15, and 16 are acidic.

Oxide	Chemical Equation	
Na ₂ O	Na ₂ O(s) + 2HCl(aq) → 2NaCl(aq) + H ₂ O(l)	
MgO	MgO(s) + 2HCl(aq) → MgCl ₂ (aq) + H ₂ O(l)	Used in indigestion remedies by neutralising the excess acid in the stomach
Al ₂ O ₃	Al ₂ O ₃ (s) + 3H ₂ SO ₄ (aq) → Al ₂ (SO ₄) ₃ (aq) + 3H ₂ O(l) Al ₂ O ₃ (s) + 2NaOH(aq) +3H ₂ O(l) → 2NaAl(OH) ₄ (aq)	Reacts with acid to form a salt and water Reacts with hot, concentrated alkali to form a salt
SiO ₂	SiO ₂ (s) + 2NaOH(aq) → Na ₂ SiO ₃ (aq) + H ₂ O(l)	Reacts with hot, concentrated alkali to form a salt and water
P ₄ O ₁₀	P ₄ O ₁₀ (s) + 12NaOH(aq) → 4Na ₃ PO ₄ (aq) + 6H ₂ O(aq)	
SO ₂	SO ₂ (g) + 2NaOH(aq) → Na ₂ SO ₃ (aq) + H ₂ O(l)	
SO ₃	SO ₃ (g) + 2NaOH(aq) → Na ₂ SO ₄ (aq) + H ₂ O(l)	

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Period 3 Hydroxides:

Hydroxide	Behavior	Reaction with Acid	Reaction with Base
NaOH	Strong base	NaOH(aq) + HCl(aq) → NaCl(aq) + H ₂ O(l)	
Mg(OH)2	Base	Mg(OH)2(s) + 2HCl(aq) → MgCl2(aq) + 2H2O(l)	

Al2(OH)3	Amphoteric	Al(OH)3(s) + 3HCl(aq) → AlCl3(s) + 3H2O(l)	Al(OH)3(s) + NaOH(aq) → NaAl(OH)4(aq)
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Physical Properties of Period 3 Chlorides:

Period 3 chloride	NaCl	MgCl ₂	Al ₂ Cl ₆	SiCl ₄	PCl ₅	SCl ₂
Chemical bonding	Ionic	Ionic	Covalent	Covalent	Covalent	Covalent
Structure	Giant ionic	Giant ionic	Simple molecular	Simple molecular	Simple molecular	Simple molecular

Reactions of Period 3 Chlorides with Water:

Chloride	Reaction	Observation
NaCl	No reaction. Hydrated ions of sodium and chloride formed as polar water molecules are attracted to the ions, dissolving the chloride and breaking down the ionic lattice.	
MgCl ₂	No reaction. Hydrated ions of magnesium and chloride formed as polar water molecules are attracted to the ions, dissolving the chloride and breaking down the ionic lattice.	
AlCl ₃	Al ₂ Cl ₆ + H ₂ O → 2[Al ₂ (H ₂ O ₆)] ³⁺ + 6Cl ⁻ [Al ₂ (H ₂ O ₆)] ³⁺ → [Al ₂ (H ₂ O) ₅ OH] ²⁺ + H ⁺ H ⁺ + Cl ⁻ → HCl [Al ₂ (H ₂ O ₆)] ³⁺ is hexaaqua aluminum ion	White fumes of HCl. The H ⁺ ion turns the solution acidic.
SiCl ₄	SiCl ₄ (l) + 2H ₂ O(l) → SiO ₂ (s) + 4HCl(g)	White fumes of HCl. White ppt of SiO ₂ . Some HCl dissolves to form acidic solution.
PCl ₅	PCl ₅ (s) + 4H ₂ O(l) → H ₃ PO ₄ (aq) + 5HCl(g)	Highly acidic solution. White fumes of HCl.

GROUP 17:

Physical Properties:

- 1) Fluorine exists as pale yellow gas.
- 2) Chlorine exists as a green or yellow gas.
- 3) Bromine exists as a brown liquid.
- 4) Iodine exists as a black solid or purple vapors.
- 5) Volatility decreases down the group as the size of the molecules and hence the instantaneous dipole-induced dipole forces between them increase.
- 6) Electronegativity decreases down the group as atomic radius increases and distance of bonding electrons from nucleus increases, making it harder for the nucleus to attract them.
- 7) Hence, strength of bond energies decreases down the group.
- 8) Fluorine is an exception because it is so small that when two atoms of fluorine get together their lone pairs get so close that they cause significant repulsion, counteracting the attraction between the bonding pair of electrons and two nuclei.
- 9) Reactivity decreases down the group.

Halogens as Oxidizing Agents:

- 1) Halogens oxidize metals by taking an electron from them.
- 2) Their ability to attract electrons (electronegativity) decreases down the group.
- 3) Hence their strength as oxidizing agents decreases down the group.
- 4) More reactive halogens can displace ions of less reactive halides from compounds or solutions by oxidizing them.

Reactions With Hydrogen:

Equation for Reaction	Description of Reaction
H ₂ (g) + F ₂ (g) → 2HF(g)	Reacts explosively even in cool, dark conditions
H ₂ (g) + Cl ₂ (g) → 2HCl(g)	Reacts explosively in sunlight
H ₂ (g) + Br ₂ (g) → 2HBr(g)	Reacts slowly on heating
H ₂ (g) + I ₂ (g) ⇌ 2HI(g)	Forms an equilibrium mixture on heating

The reactions become less vigorous down the group as reactivity decreases.

Thermal Stability of Hydrogen Halides:

- 1) Electronegativity of halogens decreases down the group.
- 2) Hence strength of bond energies decreases down the group.

- 3) Hence thermal energy required to break bonds decreases down the group.
- 4) Hence thermal stability decreases down the group.

Halide Ions as Reducing Agents:

- 1) To act as reducing agents, the halide must donate an electron.
- 2) Since ionic radius increases down the group, it becomes easier to lose an electron down the group.
- 3) Hence strength as reducing agent increases down the group.

Reaction of Halides with Aqueous Silver Ions Followed By Ammonia:

- 1) This is used to identify halide ions.
- 2) The halide is first dissolved in nitric acid. Silver nitrate solution is then added.
- 3) Equation of reaction is $\text{AgNO}_3(\text{aq}) + \text{X}^-(\text{aq}) \rightarrow \text{AgX}(\text{s}) + \text{NO}_3^-(\text{aq})$
- 4) Ionic equation is $\text{Ag}^+(\text{aq}) + \text{X}^-(\text{aq}) \rightarrow \text{AgX}(\text{s})$
- 5) Precipitate of AgX is formed. This can be identified using ammonia.
- 6) Chloride ion forms a white ppt which dissolves in dilute ammonia.
- 7) Bromide ion forms a cream ppt which dissolves in concentrated ammonia.
- 8) Iodide ion forms a pale yellow ppt which does not dissolve in ammonia.

Reaction of Halides with Aqueous Sulfuric Acid:

Halide Ion	Reaction with Concentrated Sulfuric Acid	Observations
$\text{Cl}^-(\text{aq})$	$\text{H}_2\text{SO}_4(\text{l}) + \text{NaCl}(\text{s}) \longrightarrow \text{HCl}(\text{g}) + \text{NaHSO}_4(\text{s})$	White fumes of HCl gas
$\text{Br}^-(\text{aq})$	$\text{H}_2\text{SO}_4(\text{l}) + \text{NaBr}(\text{s}) \longrightarrow \text{HBr}(\text{g}) + \text{NaHSO}_4(\text{s})$ $\text{H}_2\text{SO}_4(\text{l}) + 2\text{HBr}(\text{s}) \longrightarrow \text{Br}_2(\text{g}) + \text{SO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$	Reddish brown gas of Br_2
$\text{I}^-(\text{aq})$	$\text{H}_2\text{SO}_4(\text{l}) + \text{NaI}(\text{s}) \longrightarrow \text{HI}(\text{g}) + \text{NaHSO}_4(\text{s})$ $2\text{HI}(\text{g}) + \text{H}_2\text{SO}_4(\text{l}) \longrightarrow \text{I}_2(\text{g}) + \text{SO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ $6\text{HI}(\text{g}) + \text{H}_2\text{SO}_4(\text{l}) \longrightarrow 3\text{I}_2(\text{g}) + \text{S}(\text{s}) + 4\text{H}_2\text{O}(\text{l})$ $8\text{HI}(\text{g}) + \text{H}_2\text{SO}_4(\text{l}) \longrightarrow 4\text{I}_2(\text{g}) + \text{H}_2\text{S}(\text{s}) + 4\text{H}_2\text{O}(\text{l})$	Violet/purple vapour of I_2 Yellow solid of S Strong, bad smell of H_2S

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The Disproportionation Reactions of Chlorine:

- 1) **With cold NaOH at 15 °C:**
 - i) $\text{Cl}_2(\text{aq}) + 2\text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{NaClO}(\text{aq}) + \text{H}_2\text{O}(\text{l})$
 - ii) $\text{Cl}_2(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{Cl}^-(\text{aq}) + \text{ClO}^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$
 - iii) Oxidation of chlorine by increase in oxidation state from 0 in Cl_2 to +1 in NaClO
 - iv) Reduction of chlorine by decrease in oxidation state from 0 in Cl_2 to -1 in NaCl
- 2) **With hot NaOH at 70 °C:**
 - i) $3\text{Cl}_2(\text{aq}) + 6\text{NaOH}(\text{aq}) \rightarrow 5\text{NaCl}(\text{aq}) + \text{NaClO}_3(\text{aq}) + 3\text{H}_2\text{O}(\text{l})$
 - ii) $3\text{Cl}_2(\text{aq}) + 6\text{OH}^-(\text{aq}) \rightarrow 5\text{Cl}^-(\text{aq}) + \text{ClO}_3^-(\text{aq}) + 3\text{H}_2\text{O}(\text{l})$
 - iii) Oxidation of chlorine by increase in oxidation state from 0 in Cl_2 to +5 in NaClO_3
 - iv) Reduction of chlorine by decrease in oxidation state from 0 in Cl_2 to -1 in NaCl
- 3) **With Water:**
 - i) $\text{Cl}_2(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{HCl}(\text{aq}) + \text{HClO}(\text{aq})$
 - ii) $\text{HClO}(\text{aq}) \rightarrow \text{H}^+(\text{aq}) + \text{ClO}^-(\text{aq})$
 - iii) HClO and ClO^- act as sterilizing agents by killing bacteria

NITROGEN AND SULPHUR

Reactivity of Nitrogen:

- 1) Very unreactive element.
- 2) Two nitrogen atoms have a triple covalent bond between themselves.
- 3) This triple bond has a very high bond energy and is hence difficult to break.
- 4) Therefore nitrogen does not react with other elements except in extreme conditions.
- 5) Nitrogen is a nonpolar compound, therefore there are no electron rich or deficient areas in a nitrogen molecule which would attract another molecule and react with it.

The Basicity of Ammonia and the Brønsted Lowry Theory:

- 1) Ammonia can act as a Brønsted–Lowry base by accepting a proton (H^+) using the lone pair of electrons on the nitrogen atom to form an ammonium ion.
- 2) The ionic equation is $\text{NH}_3(\text{aq}) + \text{H}^+(\text{aq}) \rightarrow \text{NH}_4^+(\text{aq})$
- 3) An equilibrium mixture can be established in aqueous ammonia with the chemical equation $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$

4) The solution is weakly alkaline and there are more ammonia molecules than hydroxide ions. Hence ammonia is a weak base.

Preparation and Structure of Ammonium Ion:

- 1) Ammonia reacts with water. The equation is $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$.
- 2) The nitrogen atom in ammonia forms a dative bond with a hydrogen atom to form the ammonium ion.
- 3) Tetrahedral structure with bonds of equal length.

Preparation of Ammonia From Acid-Base Reactions of Ammonium:

- 1) Ammonium chloride can be heated to react with calcium hydroxide to form calcium chloride, water, and ammonia.
- 2) $2\text{NH}_4\text{Cl}(\text{aq}) + \text{Ca}(\text{OH})_2 \rightarrow \text{CaCl}_2 + \text{H}_2\text{O}(\text{l}) + 2\text{NH}_3(\text{g})$
- 3) NH_4^+ acts as an acid while OH^- acts as the base by donating and accepting a proton respectively.
- 4) This reaction can be used to test whether a solution contains ammonium ions. If it does, the gas released will turn damp red litmus paper blue.

Nitrogen Oxides:

- 1) Naturally formed only in extreme conditions such as lightning. Both nitrogen monoxide and nitrogen dioxide can be formed with equations $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}(\text{g})$ and $\text{N}_2(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$
- 2) Nitrogen oxides can be artificially produced in car engines and released to the atmosphere in the car’s exhaust fumes.
- 3) Catalytic converters reduce these oxides to nitrogen using a platinum catalyst in a reaction with the equation $2\text{CO}(\text{g}) + 2\text{NO}(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + \text{N}_2(\text{g})$

Pollutants:

Pollutant	Sources	Effects
Nitrogen oxides	Car engines, power plants	Form PAN and oxide sulfur dioxide in atmosphere Cause acid rain, damaging marine life, corroding buildings, increasing acidity of soils and rivers
Volatile organic compounds (VOCs) (unburnt hydrocarbons and their oxidized products)		Form PAN
PAN	Photochemical reactions between VOCs and nitrogen oxides in presence of sunlight	Cause photochemical smog Affects lungs and eyes Can harm plants if concentrates
Sulfur trioxide	$\text{NO}_2(\text{g}) + \text{SO}_2(\text{g}) \rightarrow \text{SO}_3(\text{g}) + \text{NO}(\text{g})$ $\text{NO}(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{NO}_2(\text{g})$ Cycle repeats SO_3 reacts with rainwater to form sulfuric acid	Cause acid rain, damaging marine life, corroding buildings, increasing acidity of soils and rivers

DEFINITIONS

First Ionization Energy: Energy required to remove 1 mole of electrons from 1 mole of gaseous atoms of an element to form 1 mole of ions.

Unified Atomic Mass Unit: One twelfth of the mass of a carbon-12 atom.

Relative Atomic Mass: Mass of an atom as compared to one twelfth the mass of a carbon-12 atom.

Relative Molecular Mass: Sum of atomic masses of all the atoms present in a compound as compared to one twelfth the mass of a carbon-12 atom.

Relative Isotopic Mass: Sum of masses of protons and neutrons of an atom as compared to one twelfth the mass of a carbon-12 atom

Relative Formula Mass: Sum of atomic masses of all the atoms present in a compound as compared to one twelfth the mass of a carbon-12 atom.

Note: Relative formula mass is used for ionic compounds, relative molecular mass is used for covalent compounds.

Mole: Amount of a substance containing 6.02×10^{23} particles of that substance.

Empirical Formula: Formula showing simplest ratio among different atoms present in a compound.

Molecular Formula: Formula showing exact numbers of different atoms present in a compound.

Electronegativity: The power of an atom to attract electrons to itself.

Ionic Bonding: Electrostatic attraction between oppositely charged ions (positive cations and negative anions).

Metallic Bonding: Electrostatic attraction between positive metal ions and delocalized electrons.

Covalent Bonding: Electrostatic attraction between the nuclei of two atoms and a shared pair of electrons.

Bond Energy: Energy required to break one mole of a particular covalent bond in the gaseous state.

Bond Length: Intermolecular distance of two covalently bonded atoms.

Standard Conditions: 298K and 101kPa with each substance in its normal physical state.

Enthalpy Change of Reaction: Enthalpy change when the reactants in the stoichiometric equation react to give the products under standard conditions.

Enthalpy Change of Formation: Enthalpy change when one mole of a compound is formed from its elements under standard conditions.

Enthalpy Change of Combustion: Enthalpy change when one mole of a substance is burnt in excess oxygen under standard conditions.

Enthalpy Change of Neutralization: Enthalpy change when one mole of water is formed by reacting an acid and an alkali under standard conditions.

Hess’s Law: Total enthalpy change in a chemical reaction is independent of the route by which the chemical reaction takes place as long as the initial and final conditions are the same.

Le Chatelier’s Principle: If a change is made to a system at dynamic equilibrium, the position of equilibrium moves to minimise this change.

Activation Energy: Minimum energy required for a collision to be effective.

Hydrocarbon: An element made up of carbon and hydrogen atoms only.